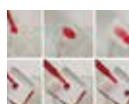




NASA satellite returns

Drifting in space for two decades, the Rossi X-ray Timing Explorer returned and burned up in the atmosphere. <https://digit.in/jun18-B35-1>



Plant inspired fake leather

Researchers have developed a self cleaning fake leather that does not get sticky, inspired by plants. <https://digit.in/jun18-B35-2>

What would you see? Let's first look at a simple math problem. You are 16 feet away from the fridge door, and you're hungrier than zombie who can smell a vat full of brains. Mathematically, it can be shown that you never quite reach the fridge.

Let's see how long it takes you to halve the distance between you and the fridge. = 16/2 feet @ 1ft/s. So $t = 8$ seconds. Now, how long does it take you to halve that distance? 4 seconds, then 1 second, then .5 seconds, then 0.25 seconds... etc. You can continue halving the distances an infinite number of times, and thus, in theory, you never reach the fridge. Time taken = $8 + 4 + 2 + 1 + 0.5 + 0.25 + 0.125 + 0.0625 + 0.03125 + 0.015625 + 0.0078125...$ etc which will total to 15.999999999999999 seconds but never quite 16 full seconds, which is the time we know it will actually take.

Yes, we're aware this is a grade five math paradox joke, we're only making it to illustrate how when we go back in time towards the Big Bang, time actually behaves like this (in the opposite direction, using the physics we know), and never quite reaches zero. Or does it?

NO BOUNDARY

This is where Hawking's theory comes in, and it's known officially as the Hartle-Hawking state. While most of us think that a before the Big Bang is an impossible state to visualise or predict because our physics doesn't work there, Hawking and James Hartle had other ideas.

Hawking explained it once as standing at the south pole, and realising that there was nothing south of the south pole. However, that didn't mean that somehow the Earth ended at the south pole! All of the hundreds of people who have stood at the south pole will testify to there being no sudden end to the Earth. It's a little bit of an anti-climax really, when you think about it. It's not like climbing Mount Everest where you can stand atop the world and objectively notice that everything is lower than you. Standing at

the south pole is literally just standing at a spot on the Earth's surface, except it's a point where the term "south" itself is meaningless, because there is no place more south on the planet.

In a similar way, Hawking and Hartle thought about the beginning of the universe as a point where time merely becomes zero by converting into imaginary time, or, perhaps into a fourth space dimension. This would allow physics to observe the Big Bang and explain how it's possible for something to come from nothing without the need for a universal singularity.

While most scientists think that the universe does indeed have a boundary



Thomas Hertog. The Belgian physicist who collaborated with Hawking.

condition, and an edge of space and time, at the moment of the Big Bang, Hawking and Hartle thought it possible that there wasn't.

As we cannot really perceive a four dimensional spacetime (3 space and 1 time dimensions), we usually dumb down to 2 dimensions of space and one of time. Thus, the popular model of the expanding universe looks a little like a cone with an constantly accelerating slope. This is the easiest way we know of to represent our constantly accelerating universe, where the speed at which galaxies are moving away from each other are directly proportional to the distance – the further away galaxies are in time, the faster they're mov-

ing away from one another. We know that the shape of the visible universe is really just a sphere centered on the observer, and spacetime is like a balloon being inflated, where each point on the balloon is moving away from every other point... and all that everyone is in agreement about... it's the apex of the cone that the Hartle-Hawking / No Boundary theory disagrees.

Hawking thought the apex of the cone was not an absolute nothingness of space or time, but was more like a badminton shuttlecock, which has half a sphere at its base, and then the cone with the ever increasing slope. Don't get us wrong, even Hawking believed that the universe started at a single point in space time, he just doesn't believe it was something special outside physics. Just as the earth doesn't begin at the north pole, the universe began at a point, but that point was nothing more than, say, the north pole of the universe.

At the very beginning of the universe, where quantum particles were popping in and out of existence, Hartle and Hawking believed that spacetime was a circular dimension like the base of the shuttlecock, but not a true singularity. Thus, before the Big Bang, the universe was indeed a singularity in space and time, but it didn't create space or time, as it had no boundaries of space or time, and merely existed in imaginary time and space.

And no, "imaginary" here is not the English usage of "made up" it's the mathematical usage which describes imaginary numbers. The number i for example, which is the square root of -1 is an imaginary number, and it's what Hartle and Hawking used to describe imaginary time in mathematical equations that this writer is far too ignorant to understand.

BACK TO THE PAPER

One of the biggest problems with the No Boundary theory was that it allowed for literally infinite universes to exist. The problem with infinity is that it allows for all possible scenarios, and thus there's a universe out there where